Audit of the SV efficiency proof, round 3: after the N1–N4 corrections

Source audited: `sv_efficiency_proof.md` (the source of `sv_efficiency_proof.tex`), rendered as `sv_efficiency_proof_md.pdf`, at the state of commit "Fix audit round-2 issues N1–N4". Rounds 1 and 2: sv_proof_audit_2026-10.md and sv_proof_audit2_2026-10.md. Not refereed.

Part A. Proof-check

A1. Scope

This round re-checks every passage changed in round 2 (H3, Remark 1.2, Lemma V, Proposition 5, Theorem 6.1). It also does a second full pass over Theorem 1 step 3, Lemma A, Lemma G, Lemma H and Corollaries 3 and 3', and spot-checks the quadrature claims against the code.

A2. Step table (changed or re-checked steps)

		$ or Friedrichs $\|w\|\|v\|\le C_F^2\varepsilon^{-1}$			w	
	convex K; $w=v-\bar v_K$, $\|w\|\le\|v\|$	gives exactly the c_F^2 of §1 (both entries)	OK (re-derived)			

A3. Verdict

- **All results are correct**, given A2 (Neumann version) and A5: Theorems 1, 2, 4 and 6, Corollaries 3 and 3', and Proposition 5. N1–N4 are closed.
- **R1** is a statement-level imprecision about quadrature. It affects no computed bound, and the code already handles it correctly.
- **R2** is a wrong description of a reference.

A4. Issues and fixes

R1 (quadrature defect: more than ε, and the wrong degree).

- **Where.** Lemma V (§11), Lemma H(iii), and the definition of $\eta_{\rm alg}$ in §10.
- **The problem.** $\delta_a^{\rm quad}:=\mathrm{Res}_A^{\rm est}(\psi_a)-\mathrm{Res}_A^{\rm asm}(\psi_a)$ is the difference between the estimator's quadrature (degree 6) and the assembly quadrature (degree 4 here, `S.degree`). That difference involves every data term tested with ψ_a: f, $b\cdot\nabla u_h$, cu_h, ε, g_N. It is not only the ε term.
- **Checked.** Constant ε, smooth problem, $\sigma=1$, converged solution: $\eta_{\rm def,A}=1.5\cdot10^{-7}$, against $2.4\cdot10^{-11}$ from the assembled residual, with $\eta_A=0.63$.
- **Consequently:**
  1. "Both defect terms vanish at the discrete solution" (Lemma H(iii)) holds only up to $\delta^{\rm quad}$.
  2. "Zero when ... ε piecewise polynomial of degree ≤ 6" should read "zero when the assembly rule integrates all data terms exactly". With the default degree-4 rule and P1 tests, that means ε of degree ≤ 4, f and b of degree ≤ 3, and c of degree ≤ 2 (since $c\,u_h\varphi$ already has degree 2 without c).
  3. "$\eta_{\rm alg}$ equals $\kappa\eta_{\rm def,A}$ when ε is integrated exactly" needs "when all data are integrated exactly".
- **Effect.**
  - Reliability: none, because the full δ_a is shifted and bounded.
  - Efficiency: $\delta^{\rm quad}$ is a data-quadrature oscillation, of the order of the quadrature error.
  - The stopping split: already right in the code, since `sv_algebraic` uses the assembled residual.
- **Fix.** Define $\delta^{\rm quad}$ once in §1 (after the defect definition) as above, and refer to it from Lemma H(iii), Proposition 5 and Lemma V.

R2 (references).

- §8 describes [AV11] (Numer. Math. 119, 2011) as "reaction–diffusion with variable coefficients". According to the abstract of [AV14], the 2011 result has a constant reaction coefficient, and [AV14] extends it to arbitrary dimension, mixed boundary conditions and a **piecewise-constant** reaction coefficient.
- [AV14] is now verified: M. Ainsworth, T. Vejchodský, CMAME 281 (2014) 184–199, doi 10.1016/j.cma.2014.08.005, arXiv 1401.2394.
- **Fix.** Correct the [AV11] description, and drop "[from memory — verify]" from [AV14] with the precise scope.

A5. Optimality (changes since round 2)

- **The c_b-free track.** The variable-weight energy norm is supported more precisely by [AV14]. A **piecewise-constant** weight $\sigma_K:=\operatorname{ess\,inf}_K(c-\frac12\mathrm{div}\,b)$ is exactly their setting, for reaction–diffusion. Fully variable reaction weights were still listed as open in a 2018 talk by Vejchodský.
- **What the track gives.** With σ_K cellwise, Remark 1.2 gives $|B_S'(w,v)|\le\max_K\{1,c_{b,K}\}|||w|||_\sigma|||v|||_\sigma$, with the local ratio $c_{b,K}=\sup_K/\inf_K(c-\frac12\mathrm{div}\,b)$. The global c_b becomes a local oscillation ratio of the reaction, which is 1 for piecewise-constant reaction.
- **Cost.** A cellwise σ in the patch problems, in Lemma G (cellwise θ_K) and in the defect solve.
- Everything else as in round 2.

Status of the fixes (18 Oct 2026)

- **R1:** fixed.
  - The quadrature defect δ^quad = Res_A^est(ψ_a) − Res_A^asm(ψ_a) is now defined once in §1, for all data terms, with the exactness conditions of the degree-4 assembly rule and osc_q := ‖ψ^quad‖_*.

  - Lemma H(iii), §9, Theorem 4.3, the split and definition of η_alg in §10, Proposition 5.2 and Lemma V now refer to it.
  - At the discrete solution the defect terms reduce to the quadrature part instead of vanishing.
- **R2:** fixed. [AV11] is described as constant reaction (2D). [AV14] is entered with its verified scope: CMAME 281 (2014) 184–199, piecewise-constant reaction, any dimension, mixed boundary conditions.

Part B. Literature scan (round 3)

B1. Answer

There is no change in novelty or track. Three findings:

1. **[AV14] verified.** It covers piecewise-constant reaction, arbitrary dimension and mixed boundary conditions; fully variable reaction is open. That makes the cellwise-σ version of the c_b-free track an adaptation of [AV14] to convection and algebraic stabilisation, not a new idea.
2. **Nothing new in 2024–2026** for equilibrated estimators with a variable-reaction weighted norm in convection–diffusion–reaction. Cai & Cai (Numer. Math. 2025) is residual/hybrid; Eigel–Merdon 2016 uses the augmented norm with constant σ.
3. **Quadrature.** I found no equilibration paper that treats the assembly-quadrature inconsistency of the compatibility explicitly. The literature handles non-polynomial data through data oscillation (e.g. arXiv 1610.01804, arXiv 2106.10945), which is how R1 should be stated.

B2. Frontier changes since round 2

result	scheme	estimate	hypotheses	mechanism	uniform in
Ainsworth–Vejchodský, CMAME 281 (2014)	conforming P1, reaction–diffusion	guaranteed, robust energy bound	**piecewise-constant** reaction; any dimension; mixed BCs	Ainsworth–Oden boundary fluxes + explicit reconstruction	reaction (all sizes)

Everything else as in rounds 1–2: Li 2026, Licht 2019, Ahmed–Jha 2026, SV20, Verfürth 2005, Cai & Cai 2025, Eigel–Merdon 2016, ESV10, ABR17, Ern–Vohralík 2013, HPSV21.

B3. Open items (ranked by novelty-to-effort)

1. **R1, R2:** text only. Low effort.
2. **The c_b-free track with a cellwise σ** (adapting [AV14]). Medium effort; the most useful extension.
3. **A5:** read Licht 2019, Theorem 1. Low effort; needs network access.
4. **The stopped iterate against the discrete solution.** Open, high risk.

B4. References (new or verified in round 3)

- M. Ainsworth, T. Vejchodský, Robust error bounds for finite element approximation of reaction–diffusion problems with non-constant reaction coefficient in arbitrary space dimension, CMAME 281 (2014) 184–199, arXiv 1401.2394; ar5iv. [abstract]
- T. Vejchodský, talks: 2015, 2018, 2011; NCMM record. [slides/abstracts]
- I. Smears, M. Vohralík, ESAIM: M2AN 54 (2020), UCL eprint. [read]
- D. Cai, Z. Cai, Numer. Math. (2025). [arXiv version read]
- M. Eigel, C. Merdon, J. Sci. Comput. 67 (2016). [read]
- Data oscillation in equilibrated estimates: arXiv 1610.01804, arXiv 1703.04987, arXiv 2106.10945, arXiv 1903.05915. [abstracts]